

Final answers (record here — graded on the answer; show your work):

1. _____ 2. _____ 3. _____ 4. _____ 5. _____ 6. _____ 7. _____
8. _____ 9. _____ 10. _____

State the domain (denominator restrictions)

Example (State the domain (denominator restrictions)):

$f(x) = \frac{1}{x+3}$: set $x+3 \neq 0$, so $x \neq -3$. Domain: all reals except -3 .

Directions: Set the denominator not equal to zero and solve.

1. $f(x) = \frac{1}{x-4}$

3. $f(x) = \frac{x}{x^2-9}$

2. $f(x) = \frac{1}{x+7}$

4. $f(x) = \frac{2}{x^2-16}$

State the domain (radical restrictions)

Example (State the domain (radical restrictions)):

$f(x) = \sqrt{x+6}$: set the radicand ≥ 0 : $x+6 \geq 0$, so $x \geq -6$.

Directions: Set the radicand ≥ 0 and solve.

5. $f(x) = \sqrt{x-1}$

7. $f(x) = \sqrt{10-x}$

6. $f(x) = \sqrt{2x-8}$

State domain AND range

Example (State domain AND range):

$f(x) = x^2 - 5$ has no root or denominator, so the domain is all reals; the lowest output is -5 , so the range is $y \geq -5$.

Directions: State the domain, then find the range by identifying the highest or lowest output.

8. $f(x) = x^2 + 7$

9. **(challenge)** $f(x) = \frac{\sqrt{x-1}}{x-4}$ — give the domain only (two restrictions apply at once)

Challenge (same sheet):

10. **(challenge)** Find the domain of $f(x) = \frac{\sqrt{x+3}}{x-1}$, showing both restrictions and how they combine.

Answer Key — fully worked

Procedure: Denominator restrictions: set the denominator $\neq 0$ and solve for the excluded x -value(s).; Radical restrictions: set the radicand ≥ 0 and solve the resulting inequality.; If a function has both a root and a denominator, apply BOTH restrictions and combine them.; For range, find the highest or lowest output the graph reaches..

1. $x \neq 4$
2. $x \neq -7$
3. $x \neq \pm 3$ (since $x^2 - 9 = 0$ at $x = 3$ and $x = -3$)
4. $x \neq \pm 4$ (since $x^2 - 16 = 0$ at $x = 4$ and $x = -4$)
5. $x \geq 1$
6. $x \geq 4$ (solve $2x - 8 \geq 0$)
7. $x \leq 10$ (solve $10 - x \geq 0$)
8. Domain: all real numbers; range: $y \geq 7$
9. Domain: $x \geq 1$ and $x \neq 4$ (the root requires $x \geq 1$; the denominator excludes $x = 4$)
10. Two restrictions apply. Radical: the radicand must be ≥ 0 , so $x+3 \geq 0$, giving $x \geq -3$. Denominator: it cannot be 0, so $x - 1 \neq 0$, giving $x \neq 1$. Combine them: start with $x \geq -3$, then remove $x = 1$. Domain: $x \geq -3$ and $x \neq 1$ (i.e., $[-3, 1) \cup (1, \infty)$).